INCH-POUND

MS25032M
28 September 2022
SUPERSEDING
MS25032L
16 December 2021

MILITARY SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, 200 AMPERES, 2 PDT, (N.O.) TYPE II, NONHERMETICALLY SEALED, MECHANICALLY INTERLOCKED

Activated after 16 December 2021 and may be used for new and existing design acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-6106.

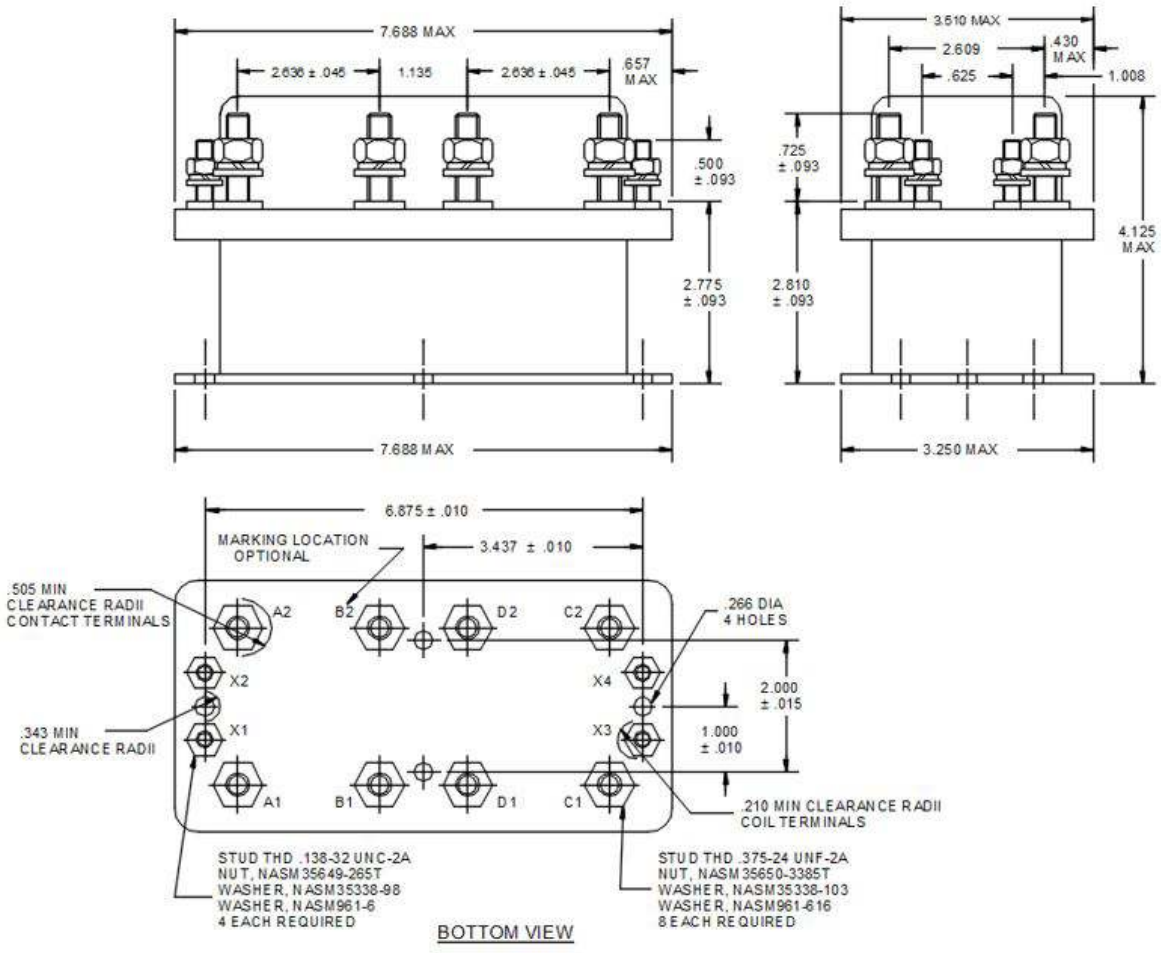


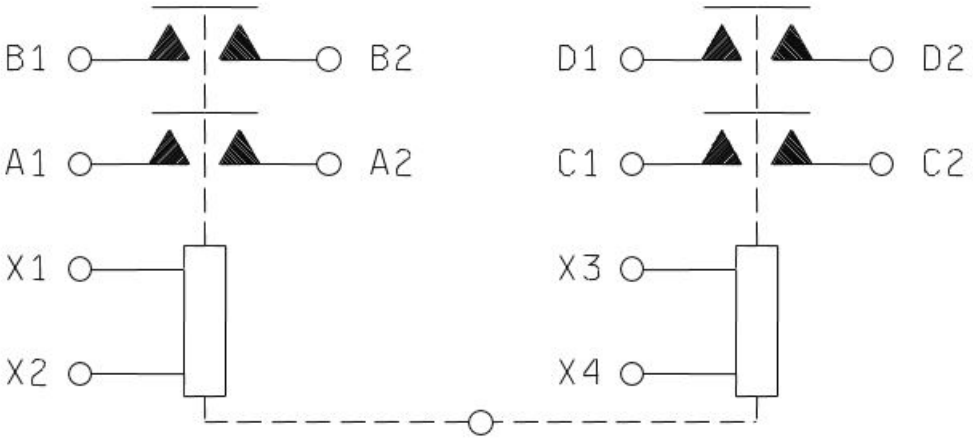
FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5945

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CIRCUIT DIAGRAM

Inches	mm	Inches	mm	Inches	mm
.010	0.25	.505	12.83	2.827	71.81
.093	2.36	.531	13.49	2.905	73.79
.138	3.51	.625	15.88	3.250	82.55
.210	5.33	.657	16.69	3.437	87.30
.266	6.76	1.000	25.40	3.468	88.09
.343	8.71	1.008	25.60	4.125	104.78
.375	9.53	1.156	29.36	6.875	174.63
.430	10.92	2.000	50.80	7.688	195.28
.500	12.70	2.609	66.27		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.032 (0.81 mm).
4. An additional flat washer may be used for terminal seating.
5. In the event of a conflict between the text of this specification and the references cited herein, this specification shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
7. Terminal numbers shall not appear on the relay header. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location.
8. Mechanical linkage shall be provided to prevent both relays from being closed at the same time.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Dash number and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 1,600 amperes.

Rupture current (NO): 2,000 amperes.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case:	1,250	N/A	1,000	N/A
Aux. contacts:	1,250	N/A	1,000	N/A
All other points:	1,250	1,500	1,000	1,125

Altitude, 1 minute.

	28 V dc	115 V ac
Coil to case:	500	N/A
Aux. contacts:	500	N/A
All other points:	500	500

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Environmental characteristics:

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet.

Shock, g-level: 25 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration levels, sinusoidal: See table IV.

Acceleration: 10 g's.

Qualification by similarity: See MIL-PRF-6106.

Part or Identifying Number (PIN): MS25032-D1.

TABLE I. Dash number and general characteristics.

PIN MS25032-	Type	Coil	Terminal type	Mounting means	Maximum weight (pounds)
D1	II	dc	Stud	Plate	5.5

TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	200				150							
Inductive	10	100											
Motor	50	150				100							
Lamp													
Transfer load 2/													
Mechanical life (reduced current)	100	50											
Mixed loads	See MIL-PRF-6106												

1/ Absence of value indicates that the relay is not rated for 3 phase applications.

2/ Transfer load indicates that the relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

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TABLE III. Operating characteristics.

PIN MS25032-	Coil data											Time - (milliseconds maximum)					
	Coil	Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Bounce			
														Main		Aux	
		Volts 1/	Freq. (Hz)	Res (Ω)	Volts	Ampere	Normal 2/	High temp test	Cont Current test					NO	NC	NO	NC
D1	X1,X2 Y1,Y2	28	dc	41	29	.80	18	21	22.5	7.0	1.5	40	15	5.0			

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

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TABLE IV. Vibration (sinusoidal).

PIN MS25032-	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
D1	.08 DA	.06 DA	2 g's	2 g's	N/A

Quality assurance provisions: Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes and shall impose additional testing requirements as necessary.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

NASM961 NASM35649 NASM35338 NASM35650
MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA-CC

Preparing activity:
DLA - CC

(Project 5945-2022-049)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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